

Two-Stage Fusion for Routing and Classification in Parameter-Efficient Continual Learning

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Abstract. Task-specific parameter-efficient continual learners maintain adapters whose selection affects both routing and classification. We augment HRM-PET with a random-projection (RP) head fitted by ridge regression over fixed first-task LoRA features. Its sufficient statistics are accumulated task by task without retaining examples. The same RP scores support two inference-time operations: task-level fusion with the initial adapted prediction before re-matching, and class-level fusion with the final logits using a top-two-margin gate. Across three benchmarks and four seeds, DRM-proposal accuracy increases by 1.10–2.26 points and classification accuracy by 0.69–1.50, with no consistent improvement in forgetting or backward transfer. Single-seed ablations across five backbones find little added routing benefit under ungated class fusion. Four-seed conditional contrasts show a positive routing contribution with gated class fusion on ImageNet-R and CUB-200, but an unresolved effect on CIFAR-100. The gate’s estimated contribution is negative on CUB-200, with a confidence interval including zero. A grid over the two mixing weights shows that classification accuracy is nearly flat in the routing weight, so the reported value is not a resolved optimum.

Keywords: Continual learning · Parameter-efficient tuning · Parameter routing · Random projection · Signal fusion.

1 Introduction

Continual learning asks a model to absorb a stream of tasks without losing what earlier ones taught it. Parameter-efficient tuning adapts a frozen pre-trained backbone using a small number of parameters. One important family maintains task-specific prompts or adapters and selects them at inference [5, 3]. Such methods can avoid storing old examples, but preserving parameters does not guarantee preserved accuracy.

In this family, task identity is unknown at inference and parameter selection is itself a prediction. A wrong adapter can produce unsuitable features. Larger pools, attention-weighted mixtures and explicit re-matching offer different responses to this difficulty [4, 7, 1]. We focus on HRM-PET, where routing and classification use related predictions from the adapted and unadapted backbone.

We add a complementary predictor: first-task adapter features, a frozen random projection and a ridge classifier fitted from accumulated statistics, following RanPAC [2]. Fitting uses current-task data without changing LoRA/TII training or retaining examples. Its scores revise the task proposal before re-matching and blend with final class logits, with greater weight when the primary classifier’s top-two margin narrows.

Contributions. We identify complementary errors in HRM-PET (Sect. 2.1) and introduce two inference-time fusion operations using one RP head: task-proposal fusion before re-matching and gated class fusion afterwards. The head is fitted in closed form without gradient descent. A three-dataset, four-seed study separates gains in routing/classification from unresolved retention effects. Conditional ablations and weight comparisons examine what each operation adds. We also evaluate the method across four datasets and five pre-training settings to examine the consistency of its gains.

2 Related Work

Parameter pools and the selection they require. L2P [4] and Dual-Prompt [5] exemplify pool-based parameter-efficient continual learning: freeze the backbone, keep a pool of lightweight parameter sets, and pick from it at inference using a query from the bare backbone. CODA-Prompt [7] replaces the discrete pick with an attention-weighted combination; HiDe-Prompt [3] separates within-task prediction, task-identity inference and task-adaptive prediction and optimises them apart, observing that identity inference, not within-task classification, is what limits the pipeline. HRM-PET [1], our baseline, substitutes LoRA adapters [6] for prompts and adds re-matching stages that revisit the initial decision. Later work enlarges the expert family: MoTE [14] combines task-specific experts, and DLEPEM [15] combines general and adapted prototypes for matching.

Closed-form solutions in continual learning. Closed-form classifiers preserve past contributions through additive statistics of fixed features. APER [8] adapts the backbone on the first task, then uses frozen features and class prototypes. ACIL [20] uses recursive least squares, replacing a sample buffer with correlation statistics and recovering the corresponding joint solution. DS-AL [21] adds a second analytic stream. RanPAC [2] applies a fixed random projection to frozen features, accumulates second-order statistics and solves a ridge system.

Our head belongs to this family and we claim no novelty in it: the projection, the Gram accumulation and the ridge solve are RanPAC’s. What differs is its role. Rather than using the solved head as the sole predictor, we retain the adapter-routing pipeline and consult the same head at two decision points.

Correcting the classifier after the fact. Class-incremental learning often biases predictions towards recent classes [9]. iCaRL [10] uses nearest exemplar means; BiC [11] learns an affine logit correction on held-out data; Weight Aligning [12] matches new- and old-class weight norms; and LUCIR [13] normalises features and weights. Our fusion leaves the adapters, auxiliary classifier and shared head fixed. It retains no exemplars but requires current-task data to fit the RP head.

2.1 HRM-PET: The Baseline We Build On

We use the released HRM-PET [1] inference path. The routing and classifier-overlap diagnostics below use seed 42 and equal task weights, as in Sect. 4.1.

Task-identity inference (TII). The input image is passed through the *bare* backbone, with no adapter attached, to obtain a query feature $q(x) = \phi(x; \theta_{\text{ptm}})$, writing ϕ for the backbone under whatever adapter it is given. That feature enters an auxiliary classifier trained with cross-entropy, which gives unnormalised class logits $u = g(q(x); \theta_\omega)$. The initial match is then $t_0 = \mathcal{T}(\arg \max_c u_c)$.

Re-matching (DRM and CRM). The first adapted pass attaches p_{t_0} and produces $r = g_{\theta_g}(\phi(x; p_{t_0}, \theta_{\text{ptm}}))$. DRM proposes the task of $\arg \max r$ and evaluates its adapter when it differs from t_0 ; our fusion modifies this proposal, not the preceding bare TII pass. CRM selects low-confidence samples from r by the baseline’s generalised-entropy threshold and compares the current logits, under the baseline log-sum-exp score, with the adapter of the second-ranked class of r ; it is confidence-triggered, not restricted to cases where the two task identities agree. Both reuse trained modules, can require further adapter forwards, and are heuristics: neither guarantees that accuracy cannot decrease.

Shared classification head. The selected parameter set is attached to the backbone, and the shared head produces a score vector over all classes seen so far, $\ell = g(\phi(x; p_{\hat{t}}, \theta_{\text{ptm}}); \theta_g) \in \mathbb{R}^{|\mathcal{Y}|}$, predicting $\hat{y} = \arg \max_c \ell_c$, where $\hat{t}$ is the identity after DRM and CRM. Classes not yet seen are masked with $-\infty$.

Two weaknesses this paper acts on. The baseline draws its routing evidence from TII and the shared head alone. To see what a second source would be worth we built the head of Sect. 3.2 on HRM-PET’s own LoRA features and used it as a router: at seed 42 on ImageNet-R, TII routing accuracy is 63.80%, the DRM proposal 77.79% and the RP head 77.12%, while the union of TII and the head reaches 82.99%. Dividing the RP-correct/TII-wrong rate by TII’s error rate gives 53.0% recovery, against 62.0% and 46.9% on CIFAR-100 and CUB-200. These ratios use task-balanced rates, not pooled image counts. The union is an oracle, not achieved routing accuracy, and it compares bare TII with RP rather than the actual Stage 1 inputs.

A better route need not become a better prediction: ℓ comes from the shared head, which spans every class seen so far, so a mis-routed sample can be classified correctly anyway. The RP head is moreover a complete classifier that a routing-only variant would discard: in the routing-only diagnostic ($w = 0.7$, $\beta = 0$) on ImageNet-R, the routed head reaches 74.31% and the projection head 70.08%, yet their union is 78.89%, the head alone contributing 4.58 task-balanced points and 2.43 and 3.87 on the other two. These measurements give the two components described next.

3 Method

3.1 Problem Statement

Class-incremental learning (CIL) aims to learn a sequence of T tasks while mitigating forgetting of earlier ones. The t -th task consists of n_t samples $\mathcal{D}_t = \{(x_i^t, y_i^t)\}_{i=1}^{n_t}$ drawn from a class set $\mathcal{C}_t$, where $x_i^t \in \mathcal{X}$ is an input image and $y_i^t \in \mathcal{Y}$ its label.

Three constraints define the setting. At step t only $\mathcal{D}_t$ is available for training. The class sets are disjoint, $\mathcal{C}_t \cap \mathcal{C}_{t'} = \emptyset$ for all $t \neq t'$, and at inference the *task identity is unknown*: the model must discriminate among all classes seen so far through a single shared classification head. Finally, this work operates in the exemplar-free setting: images from past tasks are never retained.

In the task-specific PET setting considered here [3, 1], a pre-trained backbone h_{ptm} with parameters θ_{ptm} is frozen and a lightweight parameter pool $\mathcal{P} = \{p_1, \dots, p_T\}$ is maintained, each set trained on one task; here every p_t is a LoRA adapter [6]. The shared classification head is denoted g_{θ_g} . Because the task identity is unknown at inference, the system is forced to *choose* which parameter set to attach for its primary prediction. Our fixed RP branch does not depend on this inferred task identity.

We distinguish four class-score vectors: bare TII scores u , initial adapted shared-head logits r , final re-matched logits ℓ , and RP scores s . The class-to-task map is $\mathcal{T}(\cdot)$; t_0 is the initial TII match and $\hat{t}_{\text{DRM}}$ the task proposed for direct re-matching. We write $z(\cdot)$ for per-sample standardisation with the population standard deviation floored at 10^{-6} .

3.2 The Second Source of Evidence

We fit an RP head after each task and reuse its scores in two inference-time fusion operations. HRM-PET parameters are not retrained. Figure 1 distinguishes the initial adapted logits from the bare TII scores and shows both reuse paths.

The RP head follows RanPAC [2]. Features $f \in \mathbb{R}^d$, $d = 768$, are column vectors from ViT-B/16 with the first task’s LoRA adapter (p_1 here, index 0 in code), fixed during accumulation and inference and unnormalised; changing adapters between tasks would mix incompatible spaces.

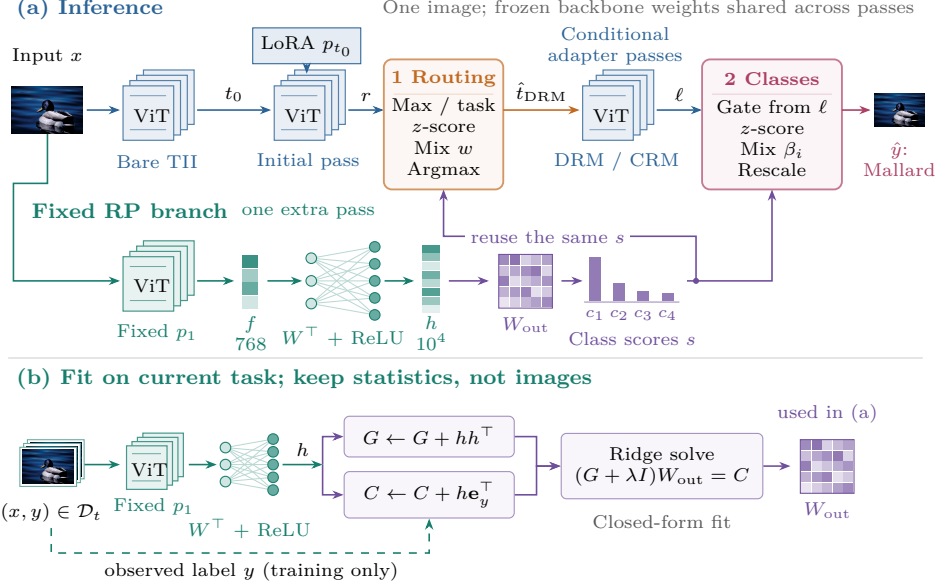


Fig. 1. Two-stage fusion. All inference passes use the same image; adapted passes share head g . TII selects t_0 ; the projection W is frozen. Stage 1 standardises task-wise maxima of r and s ; Stage 2 uses the top-two probability margin of ℓ for its continuous gate (Eqs. 6–8), then predicts by argmax. DRM runs if $\hat{t}_{\text{DRM}} \neq t_0$; CRM uses low-confidence r and its second-ranked class’s task, comparing logits by log-sum-exp (not at task 1). Only seen classes/tasks participate; score bars are schematic. Photo: T. G. Barnes/USFWS (public domain).

Projection and second-order statistics. The feature is projected into a high-dimensional space of size $M = 10^4$ and passed through a nonlinearity, and the projected features are accumulated into two matrices:

$$h = \text{ReLU}(W^\top f) \in \mathbb{R}^M, \quad W \in \mathbb{R}^{d \times M}, \quad W_{jk} \sim \mathcal{N}(0, 1), \quad (1)$$

$$G = \sum_n h_n h_n^\top \in \mathbb{R}^{M \times M}, \quad C = \sum_n h_n \mathbf{e}_{y_n}^\top \in \mathbb{R}^{M \times |\mathcal{Y}|}, \quad (2)$$

where $\mathbf{e}_y$ is the basis vector of class y . Each current-task training sample contributes once; G and C are accumulated in float64. W is generated once with seed 1993 and then frozen.

The nonlinear expansion permits richer decision functions than a linear head on f . Without ReLU, $W^\top f$ remains linear in f and adds no separating power. This does not guarantee that the expanded data become linearly separable.

Classification by ridge regression. The output weights are the closed-form solution of a regularised least-squares problem:

$$W_{\text{out}} = (G + \lambda I)^{-1} C \in \mathbb{R}^{M \times |\mathcal{Y}|}, \quad s = W_{\text{out}}^\top h \in \mathbb{R}^{|\mathcal{Y}|}. \quad (3)$$

The implementation solves the regularised system rather than inverting it.

Two inherited properties. Both G and C in (2) are *sums* over samples, so for a *fixed* feature map they, and with them W_{out} , do not depend on the order in which samples arrived: nothing an earlier task contributed is overwritten, and the head after T tasks equals the ridge solution of a joint fit over every feature seen. Their sizes do not grow with the sample count, and no per-sample feature is retained. This exact-arithmetic identity does not imply bitwise order invariance in floating point or an absence of forgetting: new classes can outscore old ones despite unchanged statistics. Moreover, changing the first task changes its adapter and hence the feature map.

3.3 Two-Stage Fusion

Both stages blend sources with different score scales. Direct mixing can make the nominal weights poorly reflect their influence. Each source is standardised per sample:

$$z(v) = \frac{v - \text{mean}(v)}{\sigma_\epsilon(v)}, \quad \sigma_\epsilon(v) = \max\{\text{std}(v), 10^{-6}\}, \quad (4)$$

where std is the population standard deviation. Moments are over seen tasks in Stage 1 and seen classes in Stage 2, excluding unseen entries.

First stage: fusion at the routing level. Stage 1 runs after the initial adapted pass and before DRM. Its input is r , the shared-head logits under the adapter selected by bare TII, not auxiliary logits u . Both r and RP scores s are mapped to seen tasks by $r_t^{\text{task}} = \max_{c \in \mathcal{C}_t} r_c$ and $s_t^{\text{task}} = \max_{c \in \mathcal{C}_t} s_c$ for $t = 1, \dots, \hat{T}$, with $\hat{T}$ the number of tasks seen so far; classes of unseen tasks never enter either maximum. The two vectors are blended after standardisation:

$$\text{fused} = w z(r^{\text{task}}) + (1 - w) z(s^{\text{task}}), \quad \hat{t}_{\text{DRM}} = \arg \max_t \text{fused}_t, \quad w = 0.7. \quad (5)$$

For one seen task the standard deviation is zero, so the floor keeps the operation finite. The fused task replaces the baseline DRM proposal $\mathcal{T}(\arg \max r)$; the initial TII match, CRM candidate rule, confidence test and shared head are unchanged. At $w = 1$ the ordering is that of r^{task} , recovering that proposal; the full baseline also needs $\beta = 0$.

Second stage: gated fusion at the classification level. Because ℓ comes from the shared head, it can predict the correct class in spite of a wrong routing decision, so we use the projection head itself as a secondary classifier, this time taking its class scores s . A fixed blend would alter confident primary predictions;

we instead reduce the RP share as the top-two probability margin grows, while recognising that confident predictions can still be wrong:

$$\beta_i = \beta \left(1 - \frac{p_{(1)} - p_{(2)}}{p_{(1)} + p_{(2)}} \right), \quad \beta = 0.5, \quad (6)$$

where $p_{(1)} \geq p_{(2)}$ are the two largest probabilities of $\text{softmax}(\ell)$ over the classes seen so far. Dividing by the sum makes the quantity depend only on the ratio $p_{(2)}/p_{(1)}$, so the gate measures how threatened the top class is rather than how much mass the pair holds. The RP share falls to zero as the top class dominates and reaches β when the two are equal: a continuous weight, not a binary test.

The blend reuses (4) and then maps the result back onto the scale of ℓ :

$$m_i = (1 - \beta_i) z(\ell_i) + \beta_i z(s_i), \quad (7)$$

$$\hat{\ell}_i = m_i \cdot \sigma_\epsilon(\ell_i) + \text{mean}(\ell_i) \in \mathbb{R}^{|\mathcal{Y}|}. \quad (8)$$

The positive affine map-back preserves rankings and restores a reference scale for cross-entropy, not a calibration guarantee: for standardised nondegenerate vectors with correlation ρ and the floor inactive, the mixture variance is $(1 - \beta_i)^2 + \beta_i^2 + 2\beta_i(1 - \beta_i)\rho$, which need not equal one. At $\beta = 0$ this stage leaves ℓ unchanged.

Cost. At inference the head costs one extra adapter forward per sample, serving both stages. That is the head’s own cost, not the pipeline’s difference, which we did not measure: changing the DRM proposal can change the number of adapter evaluations.

Per task the head accumulates G and C over the current task’s training split and solves one ridge system. Its stored arrays scale as $O(M^2 + M|\mathcal{Y}| + dM)$: G has M^2 entries, C and W_{out} each have $M|\mathcal{Y}|$, and the projection has dM . Storage does not grow with the number of observed samples for fixed dimensions and class capacity. The Gram matrix dominates in our setting. At $M = 10^4$ it holds 10^8 float64 entries, or 800 MB. Together with its regularised copy and Cholesky factor, these three matrices require approximately 2.4 GB; this is an analytical estimate, not a measured total-memory peak. A historical ten-task evaluation on ImageNet-R took 283 s on one RTX 4090 at batch 24; neither its phase-level costs nor its overhead over the baseline were measured.

4 Experiments

4.1 Setup

The backbone is ViT-B/16 [22], pre-trained on ImageNet-21K. Rank-8 LoRA modifies the key/value projections in the first five blocks. The four-seed study splits ImageNet-R [23], CIFAR-100 [24] and CUB-200-2011 [25] into 10 tasks of 20, 10 and 20 classes, respectively. The grid replaces CUB with ImageNet-A [26]

(10×20 classes) and the five-dataset protocol of [1] (one domain per task). Its pre-training settings are Sup-21K, MoCo-v3 [27], iBOT-1K/iBOT-21K [28] and DINO [29].

Each seed (42–45) has separately trained LoRA/TII parameters; pre-trained backbone weights remain frozen. Both methods use identical saved checkpoints and evaluation paths. RP fits current-task training images once using the evaluation transform: resize to 256, centre crop to 224, and convert to a tensor; CIFAR additionally uses its configured channel normalisation. No test labels enter the fit. We use $M = 10^4$, ReLU, $\lambda = 10^4$, no additional RP feature normalisation or temperature calibration, and no fusion ramp. Fusion starts at task 1; CRM retains **En=gen**, threshold -10 , and the second-ranked class candidate. Training/evaluation scripts and dataset preprocessing are versioned in the code repository [30].

Acc@task in the released evaluator measures the DRM task proposal. In the default class-based routing path, CRM can change the output logits without updating the task identity passed to this metric; it is therefore not a measure of final adapter selection after CRM. Acc@1, Acc@5 and Loss use the final classification logits. Final metrics average the per-task values with equal task weights, not by pooling all test images. Writing $a_{i,t}$ for classification accuracy on task i at stage t , the retention measures are

$$\text{Forgetting} = \frac{1}{T-1} \sum_{i=1}^{T-1} \left(\max_{i \leq t \leq T} a_{i,t} - a_{i,T} \right), \quad (9)$$

$$\text{Backward} = \frac{1}{T-1} \sum_{i=1}^{T-1} (a_{i,T} - a_{i,i}). \quad (10)$$

Both averages exclude the last task, and the maximum includes only stages at which task i has been learned. Lower is better for Loss and Forgetting.

Reproducibility and sanity check. $w = 1$ restores the baseline DRM proposal and $\beta = 0$ bypasses class fusion. With corrected unseen-class masking, this identity matches the baseline to logged precision at every stage for all three datasets and seeds 42–45; ImageNet-R seed 42 gives Acc@task 77.7914, Acc@1 74.0191 and Loss 1.2305. Re-evaluation of the main runs, controls, grids, ablations and oracle diagnostics confirmed their reported final accuracies. Some intermediate losses and retention values changed relative to earlier logs, including retention in ten full-method grid runs; we report the corrected values throughout. All forty grid evaluations were checked, and backbone/first-adapter feature tensors match across checkpoints in all main/grid checkpoint sets.

4.2 Main Results

Table 1 reports all three datasets over four seeds. The change column is pairwise — taken per seed, then averaged — so its standard deviation describes the stability of the improvement, not the seeds’ difficulty.

Table 1. Final results, mean $\pm$ sample SD over seeds 42–45; Change is paired full minus baseline. Accuracy/retention are percentages (changes in points).

Dataset	Metric	HRM-PET	Proposed	Change
<i>ImageNet-R</i>	Acc@task	77.58 $\pm$ 0.33	79.85 $\pm$ 0.18	+2.264 $\pm$ 0.158
	Acc@1	73.94 $\pm$ 0.48	75.18 $\pm$ 0.24	+1.236 $\pm$ 0.416
	Acc@5	86.71 $\pm$ 0.21	88.02 $\pm$ 0.17	+1.312 $\pm$ 0.148
	Loss	1.23 $\pm$ 0.01	1.19 $\pm$ 0.01	−0.040 $\pm$ 0.002
	Forgetting	3.36 $\pm$ 0.56	3.30 $\pm$ 0.35	−0.056 $\pm$ 0.334
	Backward	−3.16 $\pm$ 0.61	−3.20 $\pm$ 0.42	−0.047 $\pm$ 0.282
<i>CIFAR-100</i>	Acc@task	89.77 $\pm$ 0.11	90.87 $\pm$ 0.09	+1.103 $\pm$ 0.066
	Acc@1	89.78 $\pm$ 0.03	90.47 $\pm$ 0.05	+0.690 $\pm$ 0.063
	Acc@5	98.65 $\pm$ 0.03	98.86 $\pm$ 0.07	+0.218 $\pm$ 0.057
	Loss	0.39 $\pm$ 0.00	0.39 $\pm$ 0.00	−0.0003 $\pm$ 0.0012
	Forgetting	3.89 $\pm$ 0.08	3.74 $\pm$ 0.12	−0.150 $\pm$ 0.095
	Backward	−3.87 $\pm$ 0.07	−3.74 $\pm$ 0.11	+0.136 $\pm$ 0.095
<i>CUB-200</i>	Acc@task	93.12 $\pm$ 0.14	94.34 $\pm$ 0.10	+1.226 $\pm$ 0.149
	Acc@1	86.38 $\pm$ 0.25	87.88 $\pm$ 0.27	+1.495 $\pm$ 0.353
	Acc@5	97.04 $\pm$ 0.10	97.44 $\pm$ 0.13	+0.400 $\pm$ 0.103
	Loss	0.57 $\pm$ 0.00	0.53 $\pm$ 0.01	−0.036 $\pm$ 0.004
	Forgetting	2.54 $\pm$ 0.35	2.18 $\pm$ 0.12	−0.359 $\pm$ 0.325
	Backward	−2.42 $\pm$ 0.39	−2.14 $\pm$ 0.13	+0.279 $\pm$ 0.391

Intervals are paired 95% Student- t , $\bar{\Delta} \pm 3.182 s_{\Delta} / \sqrt{4}$: imprecise at four seeds, and assuming approximately normal seed-wise differences. Correlated metrics mean we discuss axes rather than count confirmations.

Routing (Acc@task): improved. The gain is +1.10 to +2.26 points, with positive paired intervals on all three datasets. This is the axis the first stage acts on directly.

Classification (Acc@1, Acc@5): improved. Acc@1 rises by +0.69 to +1.50 points and Acc@5 by +0.22 to +1.31, with positive paired intervals for both metrics on all three datasets. The full method’s Acc@1 improvement repeats across every seed. These aggregate gains do not establish how many routing repairs also repair classification.

Retention (Forgetting, Backward): unresolved. All six paired retention intervals include zero. CIFAR Forgetting is −0.150 [−0.302, +0.002] and Backward +0.136 [−0.015, +0.287]. These related metrics are not independent confirmations. Neither consistent improvement nor equivalence is established: four seeds leave substantial uncertainty, and no equivalence margin was specified. The single-seed decomposition in Sect. 4.4 is descriptive.

4.3 Placement Against Prior Methods

Table 2 places the method among the baselines the original work reports. Its first eight rows are quoted from Table 1 of [1] under Sup-21K, averages over three seeds; the last two are our own runs at seed 42 through the code path of Sect. 4.1.

Table 2. Final average accuracy (%) under Sup-21K. Rows above the rule are quoted from Table 1 of [1], averaged over three seeds; the two below are ours at seed 42, from a different implementation and seed set. Bold indicates the best value within each group, not a controlled cross-group ranking. S-Prompt++ and HiDe-LoRA are the variants evaluated in [1].

Method	CIFAR-100	ImageNet-R	ImageNet-A	5-Datasets
DualPrompt [5]	86.60	68.79	39.76	76.09
S-Prompt++ [16]	88.81	69.68	39.10	83.38
CODA-Prompt [7]	86.94	70.03	30.96	62.54
LAE [17]	85.10	72.25	40.36	64.16
CPrompt [18]	83.18	69.24	37.60	71.72
InfLoRA [19]	85.28	72.24	42.12	71.97
HiDe-LoRA [3]	88.43	71.90	42.18	93.25
HRM-PET [1]	89.45	73.86	44.28	93.99
HRM-PET, our run	89.77	74.02	45.85	91.37
Proposed	90.53	75.51	47.42	93.77

Our seed-42 baseline differs from the published mean by +0.32, +0.16, +1.57 and -2.62 points on the four datasets, respectively. These reproduction gaps preclude a controlled comparison across the two groups. The controlled comparison remains the paired baseline–full difference in Table 1.

4.4 The Full Grid

The 4×5 grid uses one code path at seed 42. All twenty cells improve Acc@task by +0.97 to +4.57 and Acc@1 by +0.59 to +3.58, against baseline Acc@1 of 31.8–93.8. Final accuracy on earlier tasks rises in every cell (+1.83 on average), versus +1.58 at learning time. Backward, their difference, worsens in five cells: learning-time accuracy rose more, not final accuracy less. This decomposition uses rounded per-task logs and remains descriptive.

4.5 Ablation

The five-backbone ImageNet-R ablation uses seed 42; baseline Acc@1 ranges from 68.47 to 74.02. We report differences between specified variants.

Routing with ungated class fusion. Routing-only and ungated class-only variants improve Acc@1 on each backbone by +0.29–+0.73 and +0.51–+3.07, respectively. Adding routing to ungated class fusion changes Acc@1 by −0.07 to +0.02, not establishing equivalence.

Full-minus-gated-class-only Acc@1 is +0.13, +0.29, +0.78, +0.43 and +0.77 on Sup-21K, MoCo-v3, iBOT-1K, iBOT-21K and DINO at seed 42. These single-seed gains establish neither reliability nor retention benefit: MoCo-v3 Forgetting rises from 3.44 to 3.69. The gate is not uniformly beneficial either: full minus ungated full is −0.26 Acc@1 on MoCo-v3 and −0.54 on iBOT-1K.

Conditional contributions over four seeds. For all three datasets and seeds 42–45, class-only fusion uses $w = 1$, $\beta = 0.5$, with or without the margin gate. Full minus gated class-only measures conditional routing; gated minus ungated class-only measures the gate without routing fusion. These are not independent stage effects.

The conditional routing effect is +0.083 [+0.026, +0.140] on ImageNet-R, −0.030 [−0.085, +0.025] on CIFAR-100 and +0.156 [+0.055, +0.257] on CUB-200. The gate is +0.603 and +0.355 on the first two, both excluding zero, and −0.284 [−0.570, +0.002] on CUB-200. Each pair sums to the full-against-ungated-class-only change, +0.69, +0.33 and −0.13, to within 0.005. Routing is positive on two datasets and unresolved on CIFAR. The gate’s CUB estimate is negative but its interval includes zero; this does not establish harm. Full still improves over HRM-PET in Table 1. Only the first fusion stage changes Acc@task.

4.6 Sensitivity to the Mixing Weights

The two weights were fixed sequentially, w before the second stage existed and β afterwards, so the second stage may have shifted the optimum of the first. Table 3 reports Acc@1 over the joint grid on ImageNet-R at seed 42 with the margin gate active.

Table 3. Acc@1 over the $w \times \beta$ grid, ImageNet-R at seed 42, full method with the margin gate.

	$\beta = 0.3$	$\beta = 0.5$	$\beta = 0.7$
$w = 0.6$	75.28	75.61	75.29
$w = 0.7$ (reported)	75.22	75.51	75.24
$w = 0.8$	75.12	75.50	75.27

The grid favours $\beta = 0.5$ at every w ; its maximum is $w = 0.6$, not the reported 0.7. At $\beta = 0.5$, Acc@1 spans 0.11 points, versus about one point for Acc@task. Across four seeds, 0.6 gains +0.332 [+0.143, +0.522] Acc@task and +0.108 [−0.064, +0.280] Acc@1 over 0.7 with the gate and $\beta = 0.5$; the Acc@1

sign reverses on seed 43. These full-method comparisons establish neither equivalence nor routing-only optimality. We retain the earlier $w = 0.7$ for consistency, not as a known optimum.

4.7 Limitations

ImageNet-R and CUB informed development; neither is an untouched holdout. More seeds do not remove selection bias. CUB/pre-training overlap [25] was not audited. The grids and five-backbone ablation are single-seed; only the three-dataset contrasts use four seeds. CIFAR’s conditional routing effect is unresolved, and gate benefits vary across backbones. Same-feature RP-only reaches mean Acc@1 69.89/88.80/87.26 on ImageNet-R/CIFAR/CUB, below the full method, but is not a same-protocol RanPAC reproduction. Quoted baselines use another implementation; our 5-Datasets result is below the published figure. Retention benefit/equivalence remains unestablished; phase-level cost and actual full-fusion repair/harm counts were not measured. The loader skips two zero-byte NotMNIST training files, also empty in the local source archive; we did not replace them.

5 Conclusion

A fixed-feature RP head supports task-proposal and class fusion, gaining 0.69–1.50 Acc@1 points across three datasets. Gated routing effects are dataset-dependent; the negative CUB gate estimate remains unresolved. Accuracy gains do not establish reduced forgetting.

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